

Ciphertext-Policy, Attribute-Based Encryption

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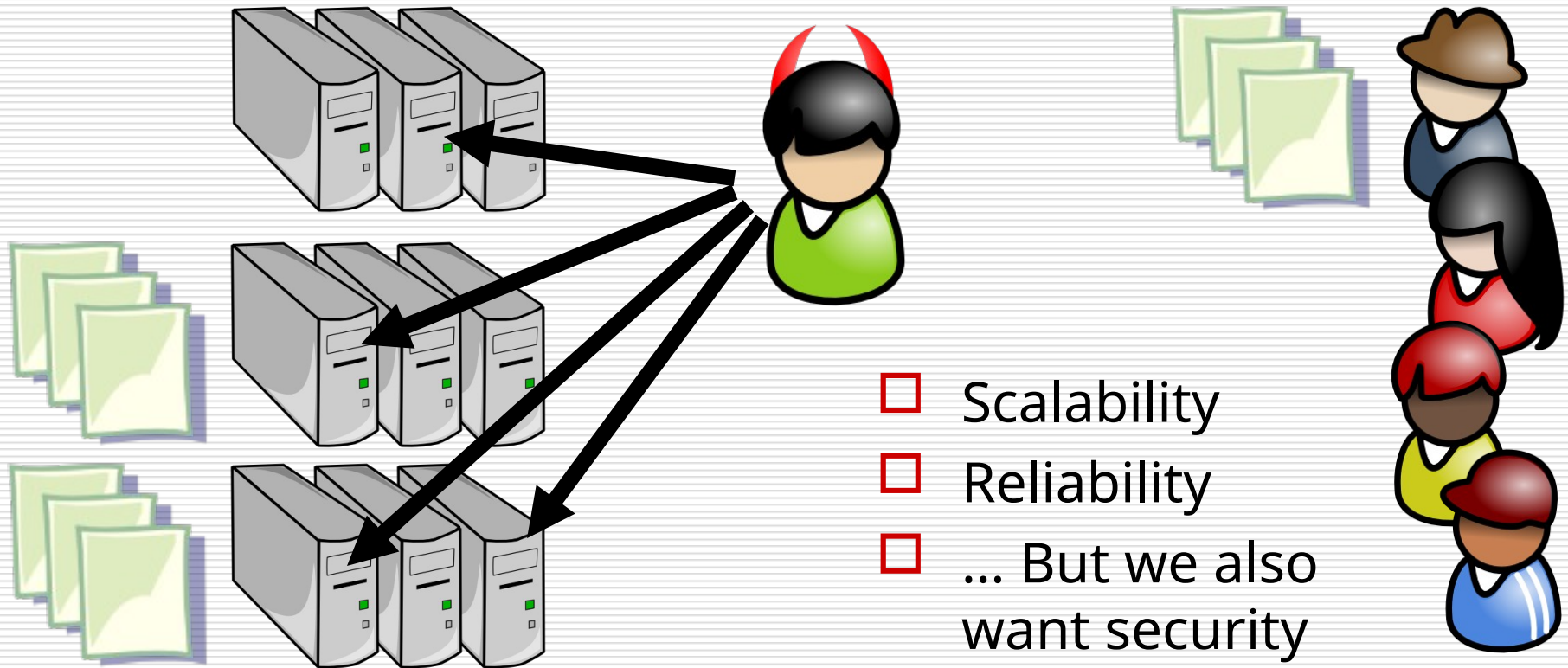
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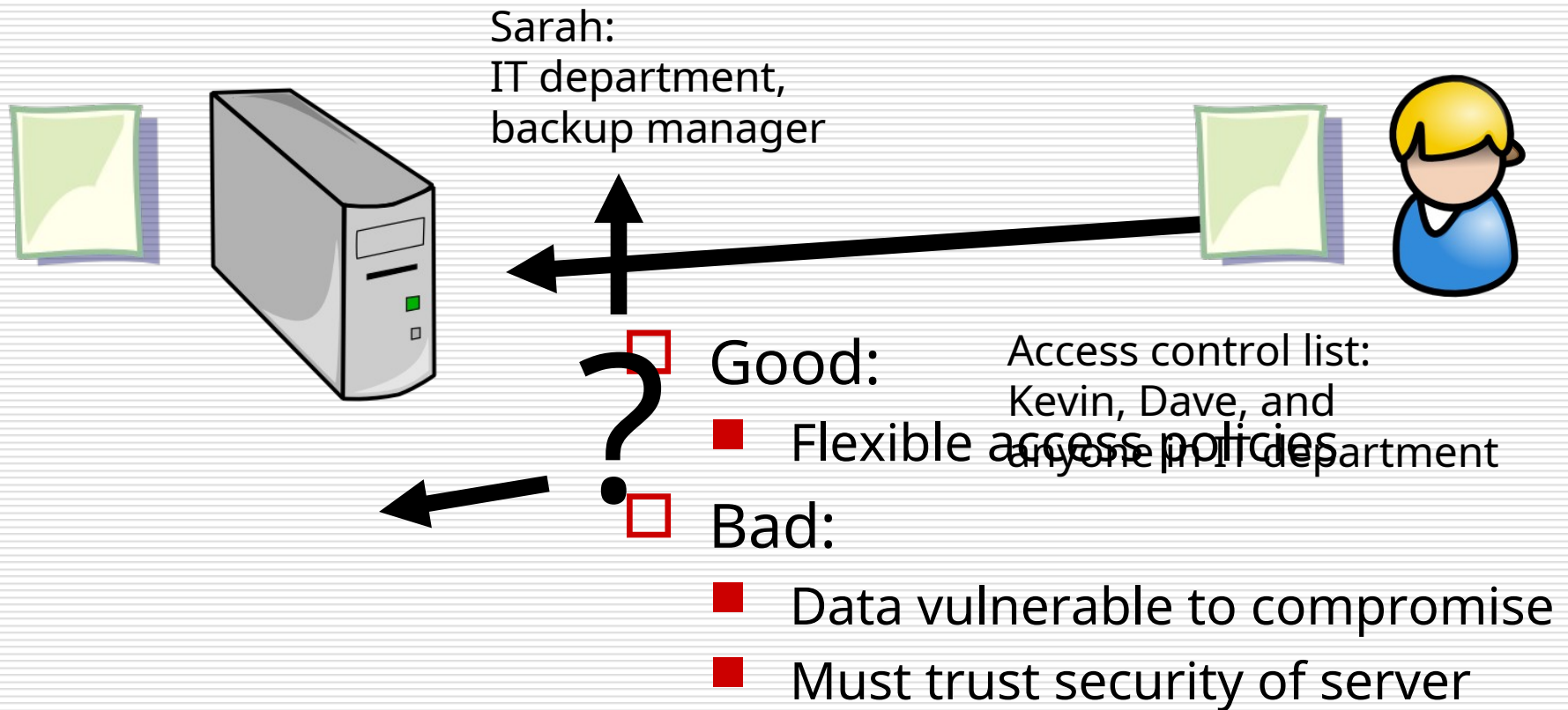
What is Ciphertext-Policy Attribute-Based Encryption (CP-ABE)?

- Type of identity-based encryption
 - One public key
 - Master private key used to make more restricted private keys
- But very expressive rules for which private keys can decrypt which ciphertexts
 - Private keys have “attributes” or labels
 - Ciphertexts have decryption policies

Remote File Storage: Interesting Challenges



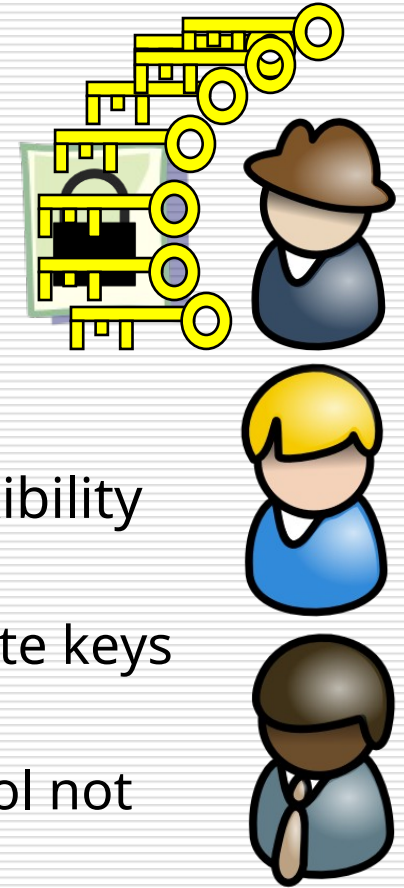
Remote File Storage: Server Mediated Access Control



Remote File Storage: Encrypting the Files



- ☐ More secure, but loss of flexibility
- ☐ New key for each file:
 - Must be online to distribute keys
- ☐ Many files with same key:
 - Fine grained access control not possible



Remote File Storage: We Want It All

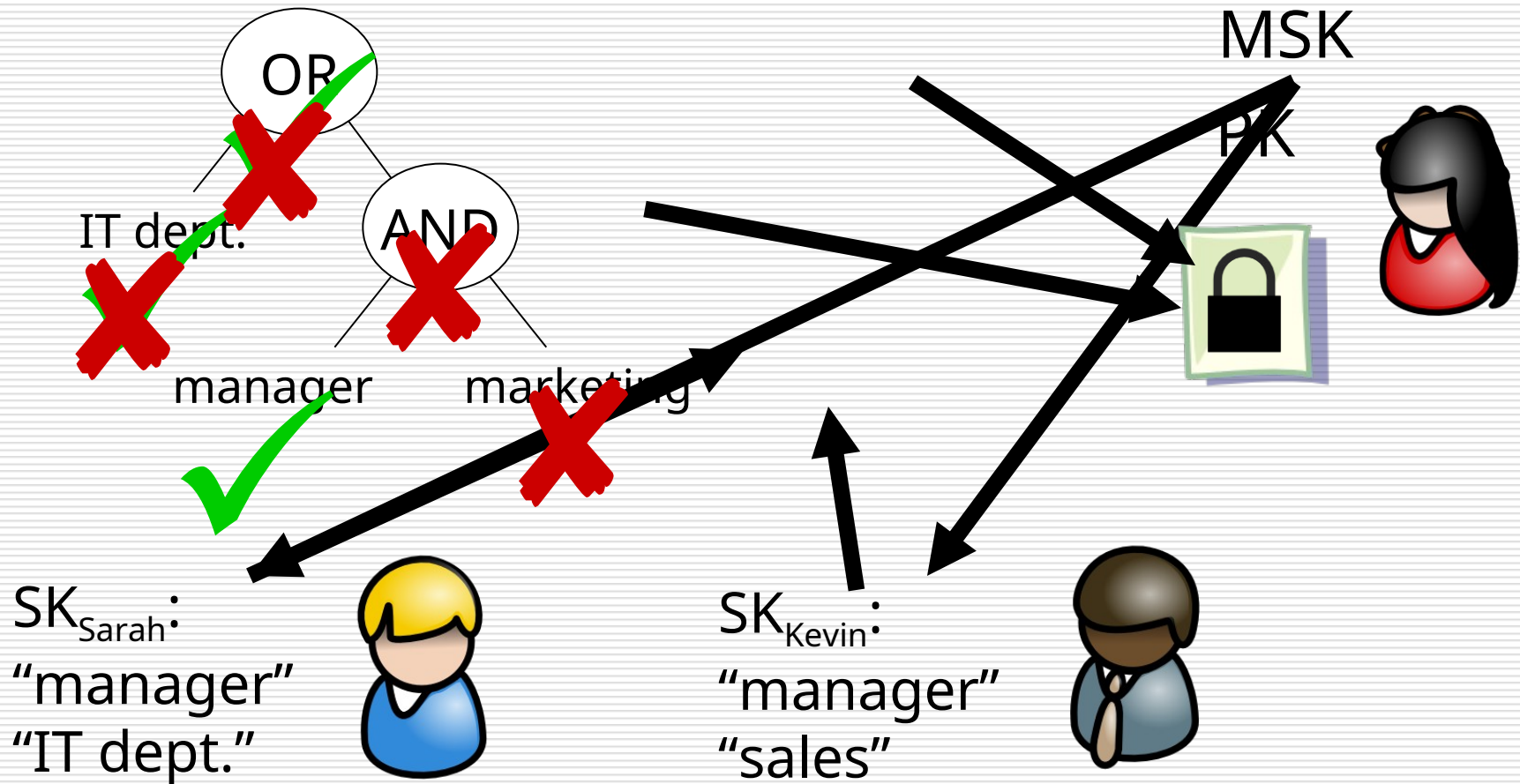
☐ Wishlist:

- Encrypted files for untrusted storage
- Setting up keys is offline
- No online, trusted party mediating access to files or keys
- Highly expressive, fine grained access policies

☐ Ciphertext-policy attribute-based encryption does this!

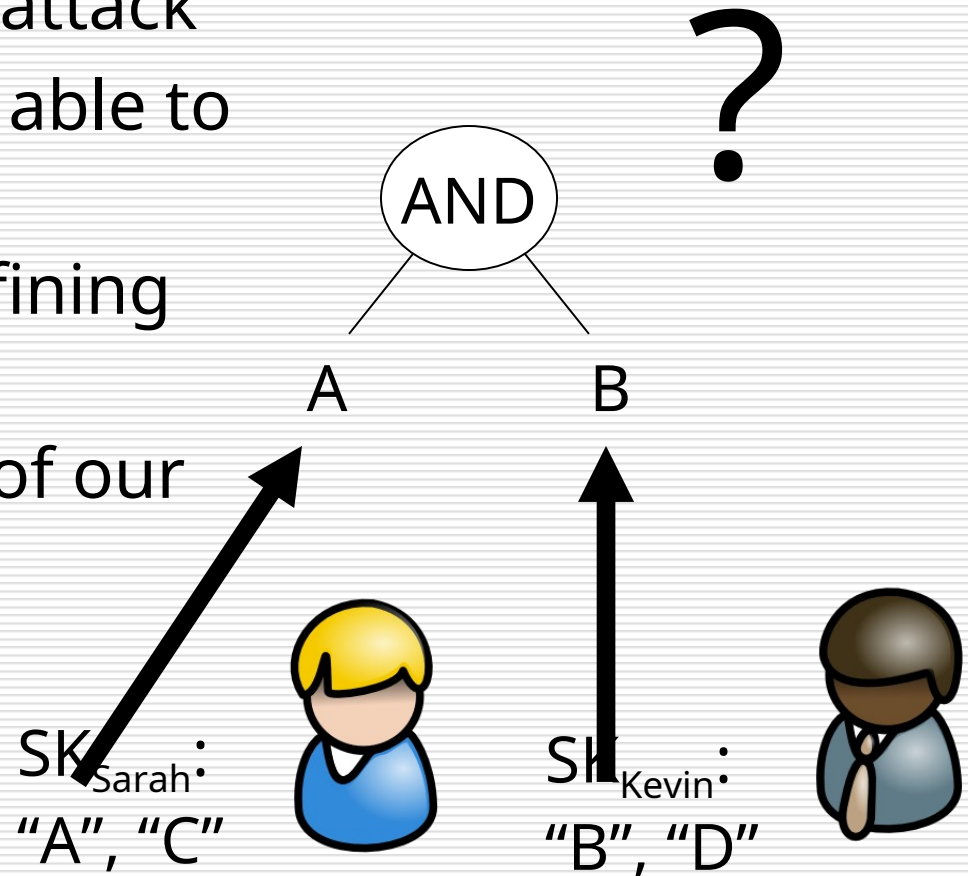
- User private keys given list of “attributes”
 - Files can encrypted under “policy” over those attributes
 - Can only decrypt if attributes satisfy policy
-

Remove File Storage: Access Control via CP-ABE



Collusion Attacks: The Key Threat

- ❑ Important potential attack
- ❑ Users should not be able to combine keys
- ❑ Essential, almost defining property of ABE
- ❑ Main technical trick of our scheme: preventing collusion



Collusion Attacks:

A Misguided Approach to CP-ABE

- ❑ Collusion attacks rule out some trivial schemes ...

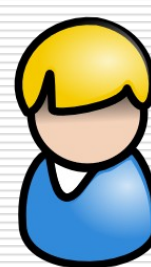
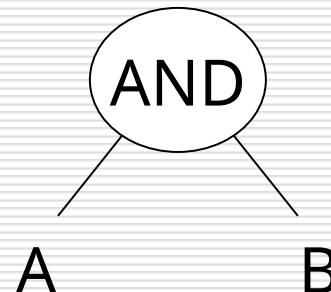
PK_A PK_B PK_C PK_D
 SK_A SK_B SK_C SK_D

$$M = M_1 + M_2$$

$$C = (E_A(M_1), E_B(M_2))$$

SK_{Sarah} :
"A", "C"

SK_{Kevin} :
"B", "D"



Highlights From Our Scheme: Background

$$|G| = |G_T| = p \qquad g \in G, \langle g \rangle = G$$

$$e : G \times G \rightarrow G_T$$

$$\forall a, b \in \mathbb{Z}^p, \quad e(g^a, g^b) = e(g, g)^{ab}$$

Highlights From Our Scheme: Public Key and Master Private Key

$$\alpha, \beta \xleftarrow{R} \mathbb{Z}^p$$

$$\text{PK} = (g, g^\beta, e(g, g)^\alpha)$$

$$\text{MSK} = (\beta, g^\alpha)$$

Highlights From Our Scheme: Private Key Generation

desired attributes: $x_1, x_2, \dots, x_n \in \{0, 1\}^*$

$$(r, r_{x_1}, r_{x_2}, \dots, r_{x_n}) \xleftarrow{R} \mathbb{Z}^p$$

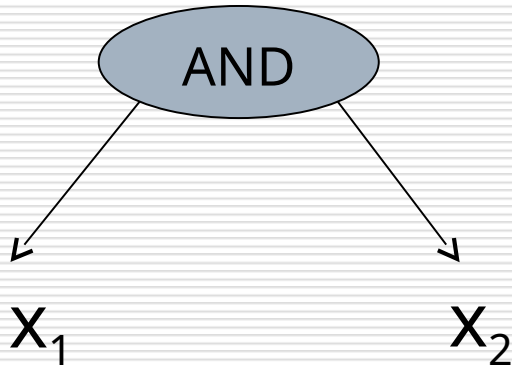
$$\text{SK} = \left(g^{(\alpha - r)/p}, \right. \\ \left. g^r H(x_1)^{r_{x_1}}, g^{r_{x_1}}, \right. \\ \vdots \\ \left. g^r H(x_n)^{r_{x_n}}, g^{r_{x_n}} \right)$$

- “Binds” key components to each other
- Makes components from different keys incompatible
- Key to preventing collusion attacks

(GB addition): Encrypt

Simpler case of AND policy

- AND policy



- $s \leftarrow \text{random}$
- Shares (AND example)

- $s_1 = s - d$

- $s_2 = d$

- For each attr:

$$C_j = g^{s_j}, H(x_j)^{s_j}$$

- For the whole message:

$$C = g^{\beta \cdot s}, \tilde{C} = Me(g, g)^{\alpha \cdot s}$$

(GB addition): Decrypt (1)

- Receive ciphertext, parse policy & attr
- For each attr user has

- Ciphertext $C_j = g^{s_j}, H(x_j)^{s_j}$
- Secret key $D_j = g^{r_{x_j}}, g^r \cdot H(x_j)^{r_{x_j}}$

- Hence:

$$\frac{e(g^r \cdot H(x_j)^{r_{x_j}}, g^{s_j})}{e(g^{r_{x_j}}, H(x_j)^{s_j})} = \frac{e(g^r, g^{s_j}) \cdot e(H(x_j)^{r_{x_j}}, g^{s_j})}{e(g^{r_{x_j}}, H(x_j)^{s_j})} = e(g, g)^{r \cdot s_j}$$

(GB addition): Decrypt (2)

- For the simpler AND policy example
(generalization very easy, now..)

- Multiply terms

$$A = \prod e(g, g)^{r s_j} = e(g, g)^{r(s-d)} \cdot e(g, g)^{r d} = e(g, g)^{r s}$$

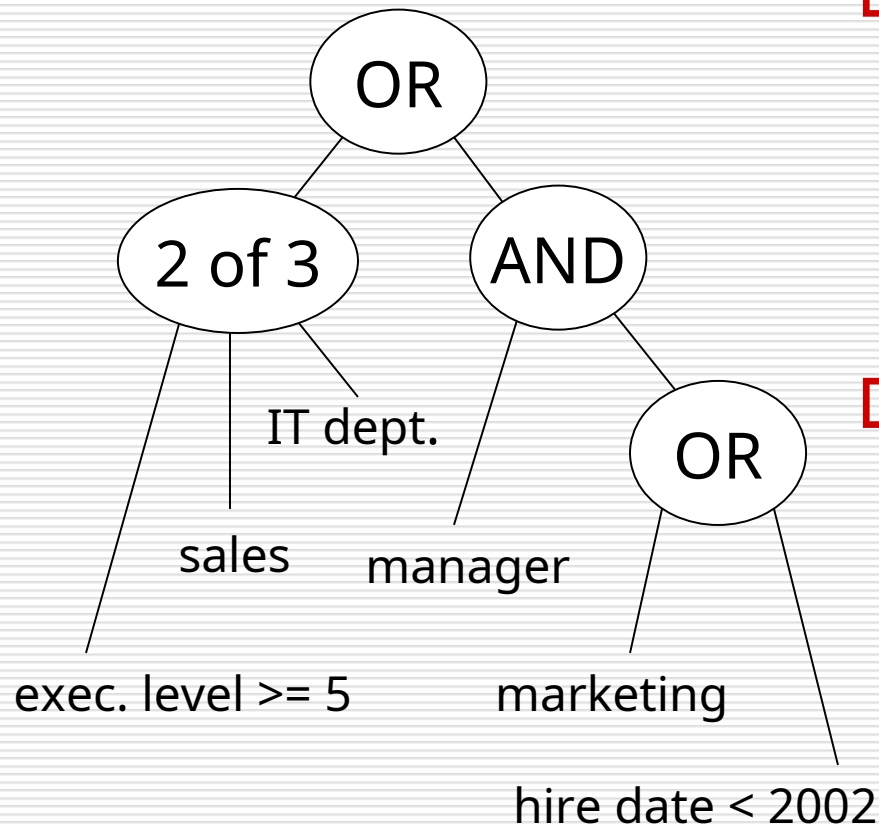
- Remember that $C = g^{\beta s}, \tilde{C} = M e(g, g)^{\alpha s}$

- And private key: $D = g^{(\alpha+r)/\beta}$

- Hence:

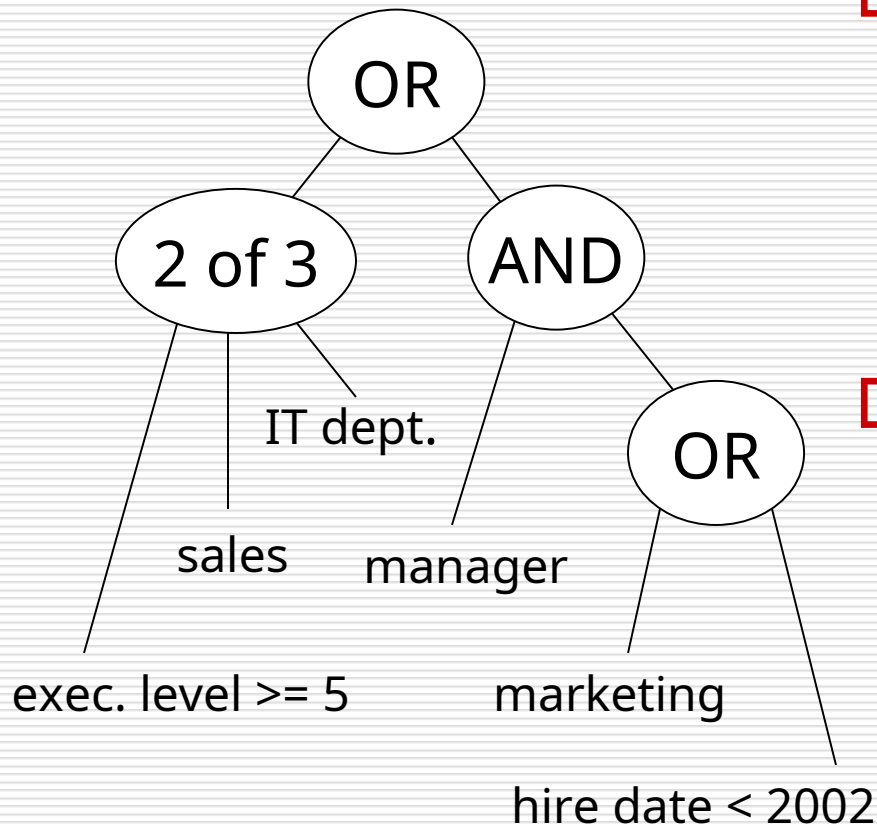
$$\begin{aligned} \frac{\tilde{C}}{e(C, D) / A} &= \frac{M e(g, g)^{\alpha s}}{e(g^{\beta s}, g^{(\alpha+r)/\beta}) / e(g, g)^{r s}} = \\ &= M \frac{e(g, g)^{\alpha s}}{e(g, g)^{\beta s(\alpha+r)/\beta} / e(g, g)^{r s}} = M \frac{e(g, g)^{\alpha s}}{e(g, g)^{s\alpha} e(g, g)^{s r} / e(g, g)^{r s}} = M \end{aligned}$$

Highlights From Our Scheme: Policy Features



- Leaf nodes:
 - Test for presence of string attribute in key
 - Also numerical attributes and comparisons
- Internal nodes:
 - AND gates
 - OR gates
 - Also k of n threshold gates

Highlights From Our Scheme: Encryption and Decryption



- Encryption:
 - Use general secret sharing techniques to model policy
 - One ciphertext component per leaf node
- Decryption:
 - Uses LaGrange interpolation “in the exponents”

Highlights From Our Scheme: Security

- Proven secure, including collusion resistance
 - Assumes random oracle model
 - Assumes generic group model
- Generic group model
 - “Black box” heuristic similar to random oracle model
 - Good future work: scheme without this assumption

Implementation: The cp-abe Toolkit

```
$ cpabe-setup
```

```
$ cpabe-keygen -o sarah_priv_key pub_key master_key \  
    sysadmin it_dept 'office = 1431' 'hire_date = 2002'
```

```
$ cpabe-enc pub_key security_report.pdf  
(sysadmin and (hire_date < 2005 or security_team)) or  
2 of (executive_level >= 5, audit_group, strategy_team))
```

Implementation: Performance

- ❑ Benchmarked on 64-bit AMD 3.7 GHz workstation
- ❑ Essentially no overhead beyond group operations in PBC library

Operation	Approximate Time
Private key gen.	35 ms per attribute
Encryption	27 ms per leaf node
Decryption	0.5–0.8 ms per leaf node

Implementation: Availability

- Available as GPL source at Advanced Crypto Software Collection (ACSC)
 - New project to bring very recent crypto to systems researchers
 - Bridge the gap between theory and practice
 - Total of 8 advanced crypto projects currently available
 - <http://acsc.csl.sri.com>

Attribute Based Encryption: Related Work

	Collusion resistant	Policies w/ infinite attr. space	Policies w/ fixed attr. space	Attributes	Policy
[1,2]	Yes	Single thresh. gate	Single thresh. gate	In ciphertext	In key
[3]	Yes	Monotone formulas	All boolean formulas	In ciphertext	In key
This	Yes	Monotone formulas	All boolean formulas	In key	In ciphertext
[4]*	No	None	All boolean formulas	In key	In ciphertext

* Has additional policy hiding property, but needs online, semi-trusted server to perform encryption

Attribute Based Encryption: Related Work

- [1] Sahai, Waters. Eurocrypt 2005.
- [2] Pirretti, Traynor, McDaniel, Waters. CCS 06.
- [3] Goyal, Pandey, Sahai, Waters. CCS 06.
- [4] Kapadia, Tsang, Smith. NDSS 07.

Thanks for Listening!

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- <http://acsc.csl.sri.com>